PES UNIVERSITY

COMPUTER COMMUNICATION NETWORK

PROJECT – REPORT

TOPIC –TICTACTOE GAME USING SOCKET PROGRAMMING

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=>>WHAT IS SOCKET PROGRAMMING:

Socket programming is a way of connecting two nodes on a network to communicate with each other. One socket(node) listens on a particular port at an IP, while the other socket reaches out to the other to form a connection. The server forms the listener socket while the client reaches out to the server.

=>> WHAT IS THREADING:

A thread is an entity within a process that can be scheduled for execution. Also, it is the smallest unit of processing that can be performed in an OS (Operating System). In simple words, a thread is a sequence of such instructions within a program that can be executed independently of other code. For simplicity, you can assume that a thread is simply a subset of a process!

=>>WHAT IS TKINTER:

Python offers multiple options for developing GUI (Graphical User Interface). Out of all the GUI methods, tkinter is the most commonly used method. It is a standard Python

interface to the Tk GUI toolkit shipped with Python. Python tkinter is the fastest and easiest way to create GUI applications. Creating a GUI using tkinter is an easy task.

WE HAVE USED THE PYTHON CODE FOR OUR PROJECT=TICTACTOE

CODE=>>

```
#Import the sockets library
import socket
import threading
import gameboard as gb
from tkinter import *
from tkinter import messagebox

import requests

def get_public_ip():
    try:
        response = requests.get('https://httpbin.org/ip')
        ip_address = response.json()['origin']
        return ip_address
    except Exception as e:
        print("Error: ", e)
        return None

# using the loopback address as my server IP address
serverAddress = '127.0.0.1'

# define a port number for my server
port = 8000

# create a socket object on my server
serverSocket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

# bind my host with my port number
serverSocket.bind((serverAddress,port))

# setup my socket using listen function
# 1 designates the max number of connections my socket
serverSocket.listen(1)
```

```
clientSocket,clientAddress = None,None

start = False
user = False
player2 = ""
board = None
pos = 0
finish = False

def createThread(con):
    thread = threading.Thread(target=con)
    thread.daemon = True
    thread.start()

def receive():
    global start,user,player2,board
    while True:
        if start and not user:
            try:
                data,addr = clientSocket.recvfrom(1024)
                name = data.decode()
                print(data)
                board = gb.Board(name)
                player2 = name
                sendData = '{}'.format("player1").encode()
                clientSocket.send(sendData)
                user = True
            except:
                pass


def receiveMove():
    global pos,start
    while True:
        if start and user:
            try:
                data,addr = clientSocket.recvfrom(1024)
                data = data.decode()
                print(data)
                if data == "Play Again":
                    reset()
```

```
            elif data == "Fun Times":
                gameOver()
            else:
                pos,board.lastMove = data.split("-")
                board.playMoveOnBoard(int(pos),board.lastMove)
                updateBoard()
                lblTurn["text"] = "You"
                play()
        except:
            pass




def acceptConnection():
    print("Thread created")
    global clientSocket,clientAddress,start
    clientSocket,clientAddress = serverSocket.accept()
    message = "Incoming play request from "+str(clientAddress)+" client?"
    answer = messagebox.askyesno("Question",message)
    if answer:
        sendData = '{}'.format("accepted").encode()
        clientSocket.send(sendData)
        start = True
        reset()
        resetStat()
        print(sendData)
        print("Client connected from: ",clientAddress)
        receive()
    else:
        sendData = '{}'.format("Not accepted").encode()
        clientSocket.send(sendData)
        clientSocket.close()
        print("Client socket closes")

def clickQuit():
    if start or user:
        serverSocket.close()
    window.destroy()

createThread(acceptConnection)
createThread(receiveMove)
```

```
window=Tk()
window.title("TiC-Tac-Toe Player1")
window.geometry("920x650+0+0")
window.configure(background = 'Cadet Blue')

tops = Frame(window, bg ='Cadet Blue', pady =2, width = 1350, height=100, relief = RIDGE)
tops.grid(row=0, column =0)

lblTitle = Label(tops, font=('arial',50,'bold'),text="Tic Tac Toe Game", bd=21, bg='Cadet Blue',fg='Cornsilk',justify = CENTER)
lblTitle.grid(row=0,column = 0)

mainFrame = Frame (window, bg = 'Powder Blue', bd=10,width = 1350, height=600, relief=RIDGE)
mainFrame.grid(row=1,column=0)

leftFrame = Frame (mainFrame ,bd=10, width =750, height=500, pady=2, padx=10, bg="Cadet Blue", relief=RIDGE)
leftFrame.pack(side=LEFT)

rightFrame = Frame (mainFrame,bd=10, width =560, height=500, padx=10, pady=2, bg="Cadet Blue", relief=RIDGE)
rightFrame.pack(side=RIGHT)

rightFrame1=Frame(rightFrame ,bd=10, width=560, height=200, padx=10, pady=2, bg="Cadet Blue", relief=RIDGE)
rightFrame1.grid(row=0, column=0)

rightFrame2 = Frame(rightFrame,bd=10, width =560, height=250, padx=10, pady=2, bg="Cadet Blue", relief=RIDGE)
rightFrame2.grid(row=1,column=0)

rightFrame3=Frame(rightFrame ,bd=10, width=560, height=150, padx=10, pady=2, bg="Cadet Blue", relief=RIDGE)
rightFrame3.grid(row=2, column=0)



def clicked1():
    global finish,board
    if start and user and btn1["text"]==" ":   #For getting the text of a button
        if board.lastMove == player2:
            btn1["text"]="O"
            board.lastMove = "player1"
            sendData = '{}-{}'.format("0",board.lastMove).encode()
            clientSocket.send(sendData)
            board.playMoveOnBoard(0,board.lastMove)
            lblTurn["text"] = "opponent"
            print(sendData)
```

```
        play()


def clicked2():
    global finish,board
    if start and user and btn2["text"]==" ":   #For getting the text of a button
        if board.lastMove == player2:
            btn2["text"]="O"
            board.lastMove = "player1"
            sendData = '{}-{}'.format("1",board.lastMove).encode()
            clientSocket.send(sendData)
            board.playMoveOnBoard(1,board.lastMove)
            lblTurn["text"] = "opponent"
            print(sendData)
            play()


def clicked3():
    global finish,board
    if start and user and btn3["text"]==" ":   #For getting the text of a button
        if board.lastMove == player2:
            btn3["text"]="O"
            board.lastMove = "player1"
            sendData = '{}-{}'.format("2",board.lastMove).encode()
            clientSocket.send(sendData)
            board.playMoveOnBoard(2,board.lastMove)
            lblTurn["text"] = "opponent"
            print(sendData)
            play()


def clicked4():
    global finish,board
    if start and user and btn4["text"]==" ":   #For getting the text of a button
        if board.lastMove == player2:
            btn4["text"]="O"
            board.lastMove = "player1"
            sendData = '{}-{}'.format("3",board.lastMove).encode()
            clientSocket.send(sendData)
            board.playMoveOnBoard(3,board.lastMove)
            lblTurn["text"] = "opponent"
            print(sendData)
            play()


def clicked5():
```

```
    global finish,board
    if start and user and btn5["text"]==" ":   #For getting the text of a button
        if board.lastMove == player2:
            btn5["text"]="O"
            board.lastMove = "player1"
            sendData = '{}-{}'.format("4",board.lastMove).encode()
            clientSocket.send(sendData)
            board.playMoveOnBoard(4,board.lastMove)
            lblTurn["text"] = "opponent"
            print(sendData)
            play()

def clicked6():
    global finish,board
    if start and user and btn6["text"]==" ":   #For getting the text of a button
        if board.lastMove == player2:
            btn6["text"]="O"
            board.lastMove = "player1"
            sendData = '{}-{}'.format("5",board.lastMove).encode()
            clientSocket.send(sendData)
            board.playMoveOnBoard(5,board.lastMove)
            lblTurn["text"] = "opponent"
            print(sendData)
            play()

def clicked7():
    global finish,board
    if start and user and btn7["text"]==" ":   #For getting the text of a button
        if board.lastMove == player2:
            btn7["text"]="O"
            board.lastMove = "player1"
            sendData = '{}-{}'.format("6",board.lastMove).encode()
            clientSocket.send(sendData)
            board.playMoveOnBoard(6,board.lastMove)
            lblTurn["text"] = "opponent"
            print(sendData)
            play()

def clicked8():
    global finish,board
    if start and user and btn8["text"]==" ":   #For getting the text of a button
        if board.lastMove == player2:
```

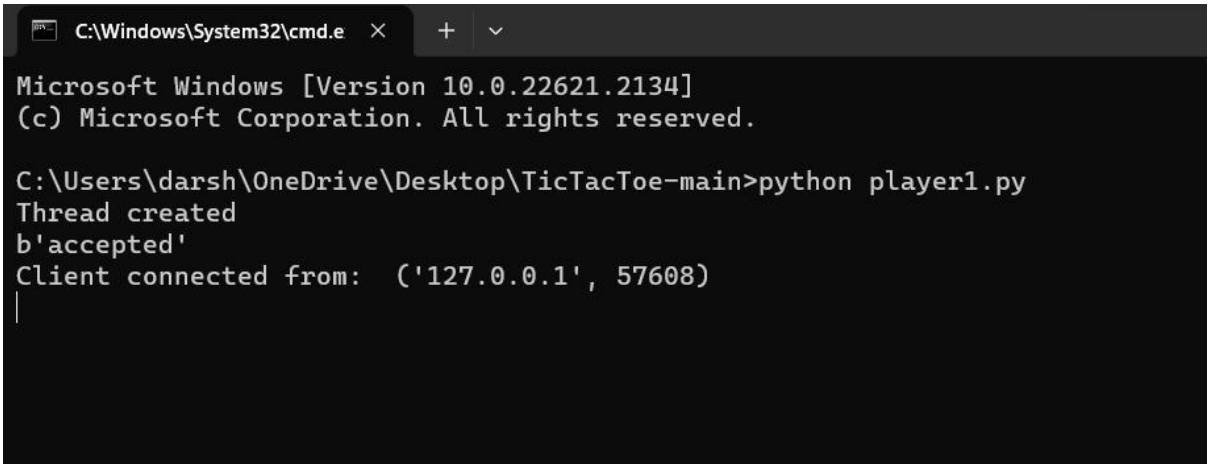
```
        btn8["text"]="O"
        board.lastMove = "player1"
        sendData = '{}-{}'.format("7",board.lastMove).encode()
        clientSocket.send(sendData)
        board.playMoveOnBoard(7,board.lastMove)
        lblTurn["text"] = "opponent"
        print(sendData)
        play()

def clicked9():
    global finish,board
    if start and user and btn9["text"]==" ":   #For getting the text of a button
      if board.lastMove == player2:
        btn9["text"]="O"
        board.lastMove = "player1"
        sendData = '{}-{}'.format("8",board.lastMove).encode()
        clientSocket.send(sendData)
        board.playMoveOnBoard(8,board.lastMove)
        lblTurn["text"] = "opponent"
        print(sendData)
        play()


def play():
    finish,won = board.isGameFinished()
    if finish:
      board.recordGamePlayed()
      board.resetGameBoard()
      board.lastMove = "player1"

def gameOver():
    isCon = False
    start = False
    user = False
    close = True
    clientSocket.close()
    board.numGames = board.numGames - 1
    comData = board.computeStats()
    lblGames["text"] = comData["numGames"]
    lblWon["text"] = comData["wins"]["O"]
    lblLost["text"] = comData["loss"]["O"]
    lblTies["text"] = comData["ties"]
```

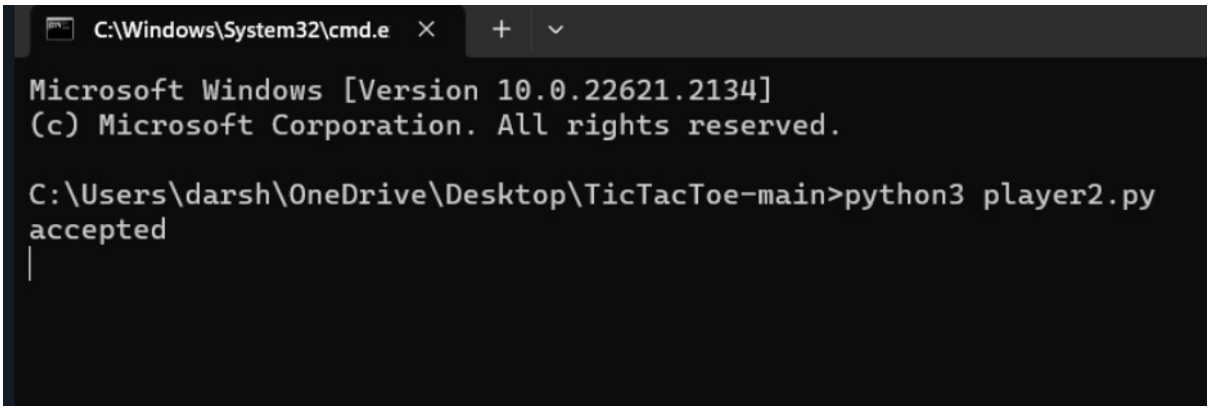
```
    lblTurn["text"] = "opponent"
    reset()

def reset():
    btn1["text"]=" "
    btn2["text"]=" "
    btn3["text"]=" "
    btn4["text"]=" "
    btn5["text"]=" "
    btn6["text"]=" "
    btn7["text"]=" "
    btn8["text"]=" "
    btn9["text"]=" "


def resetStat():
    lblTurn["text"] = "opponent"
    lblGames["text"] = " "
    lblWon["text"] = " "
    lblLost["text"] = " "
    lblTies["text"] = " "

def updateBoard():
    if int(pos) == 0:
        btn1["text"]="X"
    elif int(pos) == 1:
        btn2["text"]="X"
    elif int(pos) == 2:
        btn3["text"]="X"
    elif int(pos) == 3:
        btn4["text"]="X"
    elif int(pos) == 4:
        btn5["text"]="X"
    elif int(pos) == 5:
        btn6["text"]="X"
    elif int(pos) == 6:
        btn7["text"]="X"
    elif int(pos) == 7:
        btn8["text"]="X"
    elif int(pos) == 8:
        btn9["text"]="X"
```

```
btn1 = Button(leftFrame, text=" ",bg="yellow", fg="Black",width=8,height=3,font=('Times', 26),command=clicked1)
btn1.grid(column=1, row=1, sticky = S+N+E+W)
btn2 = Button(leftFrame, text=" ",bg="yellow", fg="Black",width=8,height=3,font=('Times', 26), command=clicked2)
btn2.grid(column=2, row=1, sticky = S+N+E+W)
btn3 = Button(leftFrame, text=" ",bg="yellow", fg="Black",width=8,height=3,font=('Times', 26),command=clicked3)
btn3.grid(column=3, row=1, sticky = S+N+E+W)
btn4 = Button(leftFrame, text=" ",bg="yellow", fg="Black",width=8,height=3,font=('Times', 26),command=clicked4)
btn4.grid(column=1, row=2, sticky = S+N+E+W)
btn5 = Button(leftFrame, text=" ",bg="yellow", fg="Black",width=8,height=3,font=('Times', 26),command=clicked5)
btn5.grid(column=2, row=2, sticky = S+N+E+W)
btn6 = Button(leftFrame, text=" ",bg="yellow", fg="Black",width=8,height=3,font=('Times', 26),command=clicked6)
btn6.grid(column=3, row=2, sticky = S+N+E+W)
btn7 = Button(leftFrame, text=" ",bg="yellow", fg="Black",width=8,height=3,font=('Times', 26),command=clicked7)
btn7.grid(column=1, row=3, sticky = S+N+E+W)
btn8 = Button(leftFrame, text=" ",bg="yellow", fg="Black",width=8,height=3,font=('Times', 26),command=clicked8)
btn8.grid(column=2, row=3, sticky = S+N+E+W)
btn9 = Button(leftFrame, text=" ",bg="yellow", fg="Black",width=8,height=3,font=('Times', 26),command=clicked9)
btn9.grid(column=3, row=3, sticky = S+N+E+W)


btnQuit=Button (rightFrame3, text="Quit", font=('arial', 20, 'bold'), height = 1, width =15,command = clickQuit)
btnQuit.grid(row=2, column=1 ,padx=6, pady=10)

lbl=Label(rightFrame2, font=('arial', 20, 'bold'), text="Games:",padx=2, pady=2, bg="Cadet Blue")
lbl.grid (row=0, column=0, sticky=W)
lblGames=Label(rightFrame2, font=('arial', 20, 'bold'), text=" ",padx=2, pady=2, bg="Cadet Blue",width=8)
lblGames.grid (row=0, column=1, sticky=W)
lbl=Label(rightFrame2, font=('arial', 20, 'bold'), text="Won:",padx=2, pady=2, bg="Cadet Blue")
lbl.grid (row=1, column=0, sticky=W)
lblWon=Label(rightFrame2, font=('arial', 20, 'bold'), text=" ",padx=2, pady=2, bg="Cadet Blue",width=8)
lblWon.grid (row=1, column=1, sticky=W)
lbl=Label(rightFrame2, font=('arial', 20, 'bold'), text="Lost:",padx=2, pady=2, bg="Cadet Blue")
lbl.grid (row=2, column=0, sticky=W)
lblLost=Label(rightFrame2, font=('arial', 20, 'bold'), text=" ",padx=2, pady=2, bg="Cadet Blue",width=8)
lblLost.grid (row=2, column=1, sticky=W)
lbl=Label(rightFrame2, font=('arial', 20, 'bold'), text="Ties:",padx=2, pady=2, bg="Cadet Blue")
lbl.grid (row=3, column=0, sticky=W)
lblTies=Label(rightFrame2, font=('arial', 20, 'bold'), text=" ",padx=2, pady=2, bg="Cadet Blue",width=8)
lblTies.grid (row=3, column=1, sticky=W)

lbl=Label(rightFrame1, font=('arial', 20, 'bold'), text="Host:",padx=2, pady=2, bg="Cadet Blue")
```

```
lbl.grid (row=0, column=0, sticky=W)

lblUsername=Label(rightFrame1, font=('arial', 20, 'bold'), text="player1",padx=2, pady=2, bg="Cadet Blue",width=10)

lblUsername.grid (row=0, column=1, sticky=W)

lbl=Label(rightFrame1, font=('arial', 20, 'bold'), text="Turn: ",padx=2, pady=2, bg="Cadet Blue")

lbl.grid (row=1, column=0, sticky=W)

lblTurn=Label(rightFrame1, font=('arial', 20, 'bold'), text="opponent",padx=2, pady=2, bg="Cadet Blue",width=10)

lblTurn.grid (row=1, column=1, sticky=W)


window.mainloop()
```

TERMINAL SCREEN SHOTS





TIC TAC TOE GAME ANALYSIS:

WIRESHARK ANALYSIS:

BEFORE THE GAME START

(ip.addr eq 127.0.0.1 and ip.addr eq 127.0.0.1) and (tcp.port eq 57608 and tcp.port eq 8000)

No.	Time	Source	Destination	Protocol	Length	Info
30	0.000000	127.0.0.1	127.0.0.1	TCP	56	57608 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
31	0.000033	127.0.0.1	127.0.0.1	TCP	56	8000 → 57608 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
32	0.000053	127.0.0.1	127.0.0.1	TCP	44	57608 → 8000 [ACK] Seq=1 Ack=1 Win=327424 Len=0
33	14.132248	127.0.0.1	127.0.0.1	TCP	52	8000 → 57608 [PSH, ACK] Seq=1 Ack=1 Win=2161152 Len=8
34	0.000047	127.0.0.1	127.0.0.1	TCP	44	57608 → 8000 [ACK] Seq=1 Ack=9 Win=327424 Len=0

AFTER THE GAME START

Apply a display filter ... <Ctrl-/>

o.	Time	Source	Destination	Protocol	Length	Info
29	3.006078	192.168.232.1	192.168.232.1	ICMP	128	Destination unreachable (Host unreachable)
30	15.503349	192.168.175.1	224.0.0.251	MDNS	75	Standard query 0x0000 PTR _microsoft_mcc._tcp.local, "QU" question
31	0.002919	192.168.232.1	224.0.0.251	MDNS	75	Standard query 0x0000 PTR _microsoft_mcc._tcp.local, "QU" question
32	0.001612	10.5.25.18	224.0.0.251	MDNS	75	Standard query 0x0000 PTR _microsoft_mcc._tcp.local, "QU" question
33	0.001181	fe80::7d13:b47a:6bd…	ff02::fb	MDNS	95	Standard query 0x0000 PTR _microsoft_mcc._tcp.local, "QU" question
34	0.000241	fe80::adec:9604:1d7…	ff02::fb	MDNS	95	Standard query 0x0000 PTR _microsoft_mcc._tcp.local, "QU" question
35	0.000420	fe80::92bc:3104:726…	ff02::fb	MDNS	95	Standard query 0x0000 PTR _microsoft_mcc._tcp.local, "QU" question
36	0.432288	127.0.0.1	127.0.0.1	TCP	56	58057 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
37	0.000065	127.0.0.1	127.0.0.1	TCP	56	8000 → 58057 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
38	0.000072	127.0.0.1	127.0.0.1	TCP	44	58057 → 8000 [ACK] Seq=1 Ack=1 Win=327424 Len=0
39	0.566940	192.168.175.1	224.0.0.251	MDNS	75	Standard query 0x0000 PTR _microsoft_mcc._tcp.local, "QM" question
40	0.000546	192.168.232.1	224.0.0.251	MDNS	75	Standard query 0x0000 PTR _microsoft_mcc._tcp.local, "QM" question
41	0.000630	10.5.25.18	224.0.0.251	MDNS	75	Standard query 0x0000 PTR _microsoft_mcc._tcp.local, "QM" question
42	0.000532	fe80::7d13:b47a:6bd…	ff02::fb	MDNS	95	Standard query 0x0000 PTR _microsoft_mcc._tcp.local, "QM" question
43	0.000207	fe80::adec:9604:1d7…	ff02::fb	MDNS	95	Standard query 0x0000 PTR _microsoft_mcc._tcp.local, "QM" question
44	0.000126	fe80::92bc:3104:726…	ff02::fb	MDNS	95	Standard query 0x0000 PTR _microsoft_mcc._tcp.local, "QM" question

RESULT:

By Using the python and socket programming we have created the tic tac toe game .and we analysed packet sharing by wire shark

THANK YOU